Algorithms: exact instructions

Programming · Level 1 · Lesson 1



Today we will

- Say what an algorithm is.
- Explain why a computer needs exact instructions.
- Play the human robot and write its program.

Remember

- An algorithm: exact steps, in order, that finish.
- A computer does what you wrote, not what you meant.
- Forward, turn left, turn right: three cards are enough.

The human robot

1. Warm-up (5 min): Describe a square without saying “square”.
2. Teach (5 min): What an algorithm is; the three cards.
3. Play (15 min): Pairs program a robot pupil on a chalk grid.
4. Debug (10 min): Find the card that went wrong, and fix it.
5. Wrap-up (5 min): Why must steps be exact? Exit ticket.

No device is needed. If you have one phone, show the robot app at the end as “the same game, on a screen”.

Let’s discuss

1. Which instruction caused the most trouble, and why?
2. Where do pupils already follow algorithms in other subjects?
3. Why can’t a computer guess what we meant?

Exit ticket

- Write an exact algorithm for one thing you did this morning.